

Structural analysis of poly(ethylene terephthalate) modified by polypropylene-graft-maleic anhydride from rheological data

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ABSTRACT: During some investigations on compatibilizing poly(ethylene terephthalate) (PET) blends by maleic anhydride-grafted polyolefins (PO-g-MA), the reaction between PET and PO-g-MA and the formation of PO-g-PET were proposed. However, the effect of PO-g-MA on the rheological properties of PET has not been the aim of any specific study. In our study, polypropylene (PP)-g-MA was melt reacted with PET, and the effects of its content and grafting degree on the intrinsic viscosity and linear viscoelastic properties of PET were investigated. The results reveal that the addition of PP-g-MA leads to increasing intrinsic viscosity, which confirms the reaction between PP-g-MA and PET. Rheological measurements suggest that the addition of PP-g-MA improves the elastic behavior and increases the complex viscosity of PET. No shear thinning behavior was observed for any samples, indicating that branching is not widespread. The increase in PP-g-MA content or grafting degree leads to increases in the intrinsic and complex viscosities and improves the elastic behavior. Furthermore, we reported the unusual observation of an increasing viscosity with frequency in modified samples due to the transesterification phenomenon. © 2018 Wiley Periodicals, Inc. *J. Appl. Polym. Sci.* **2018**, 135, 46896.

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INTRODUCTION

The low melt strengths of poly(ethylene terephthalate) (PET) and postconsumer PET (POSTC-PET) cause fundamental problems in processing this polymer, especially in processes in which a high value for melt strength is required (e.g., thermoforming, blow molding, film blowing, and extrusion foaming).^{1–6} The main reasons for low melt strength of PET and POSTC-PET are degradation reactions that take place mainly during the polycondensation stage of polymerization at 280–300 °C (leading to a low degree of polymerization),² and in melt processing at 260–290 °C.^{1–4} Figures 1 and 2 show thermal-oxidative degradation and hydrolysis of PET chains, respectively. The thermal-oxidative degradation (Figure 1) leads to breaking of macromolecular chains and consequently the formation of volatile products (like acetaldehyde), linear and cyclic oligomers, and shorter macromolecules with acid carboxylic and vinyl ester end groups. The similar mechanism in hydrolytic degradation (Figure 2) results in the formation of short chains with carboxylic acid and hydroxyl ester end groups.¹

In order to enhance the melt strength of PET, solid state polymerization (SSP) and chain extension by reactive melt-processing have been used frequently.⁴ Chain extension is a more attractive choice because it is less expensive than SSP and can be applied to an existing extrusion process. There are various functional groups that can react with the hydroxyl or carboxylic groups of PET. The most widely used functional groups are isocyanates,^{7–9} phosphites,^{10,11} anhydrides,^{12–15} and epoxides.^{16–18} Differences in nature, reactivity, and number of functional groups of the chain extender molecule lead to different linear or branched molecular structures,¹ as shown in Figure 3.

In recent years, multifunctional oligomers and copolymers containing functional groups have been used as PET chain extenders. Xanthos and Young¹⁷ used poly(ethylene-co-glycidyl methacrylate) and a diepoxide as PET chain extenders. They found complex degradation and extension reactions. Many studies concentrated on the effect of oligomeric multifunctional epoxides (for example, Joncryl ADR) on the molecular structure and rheological properties of PET.^{19–26} Raffa *et al.*²³

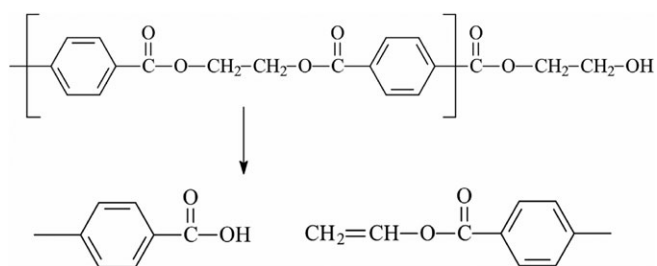


Figure 1. PET thermal degradation mechanism.¹

studied the effect of bifunctional small-molecule chain extenders and multifunctional oligomers (Joncryl ADR) on the molecular structure and rheological properties of PET. The results indicated that even with equal amounts of functional groups, samples modified by Joncryl ADR show a higher viscosity than those modified by bifunctional chain extenders. Vilalobos *et al.*¹⁹ used a similar oligomeric agent in order to branch four different grades of PET with different intrinsic viscosities. They observed a significant intrinsic viscosity increase and a broader processing window compared to regular chain extenders with low molecular weight. Xiao *et al.*²¹ and Makkam and Harnnarongchai²² used Joncryl ADR to branch out PET molecules, and they found a molecular weight increase, broadening of molecular weight and relaxation time distributions, deviation from the Cole–Cole diagram of linear PET, gelation, and an increase in extensional viscosity with the increase in chain extender concentration.

Anhydride functional groups show high reactivity with the hydroxyl groups of PET. Colin and Verdu²⁷ proposed that the anhydrides formed during the oxidation of PET can react in transesterification processes, and in certain conditions, they can even form branches. Small molecules containing anhydride groups, like pyromellitic dianhydride (PMDA), have been frequently used as PET chain extenders. In this regard, oligomers and macromolecules with anhydride groups in their molecular structure can potentially be used as PET chain extenders. Smith and Trevitt²⁸ used a styrene–maleic anhydride (SMA) copolymer in the concentration range of 0.05–0.5 wt % as PET chain extender to increase its melt strength, especially in blow molding and foaming processes. They also used PMDA as a chain extender and reported a higher melt strength in SMA-modified samples compared to PMDA-modified ones in the processes mentioned above, although intrinsic viscosity measurements showed contrary results. It should be noted that they relied upon

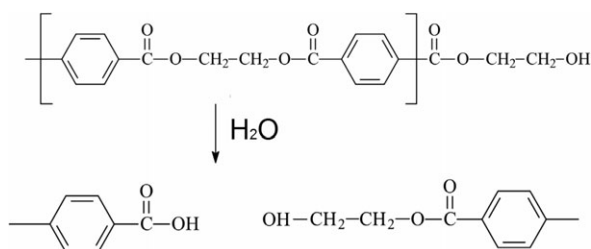


Figure 2. PET hydrolysis mechanism.¹

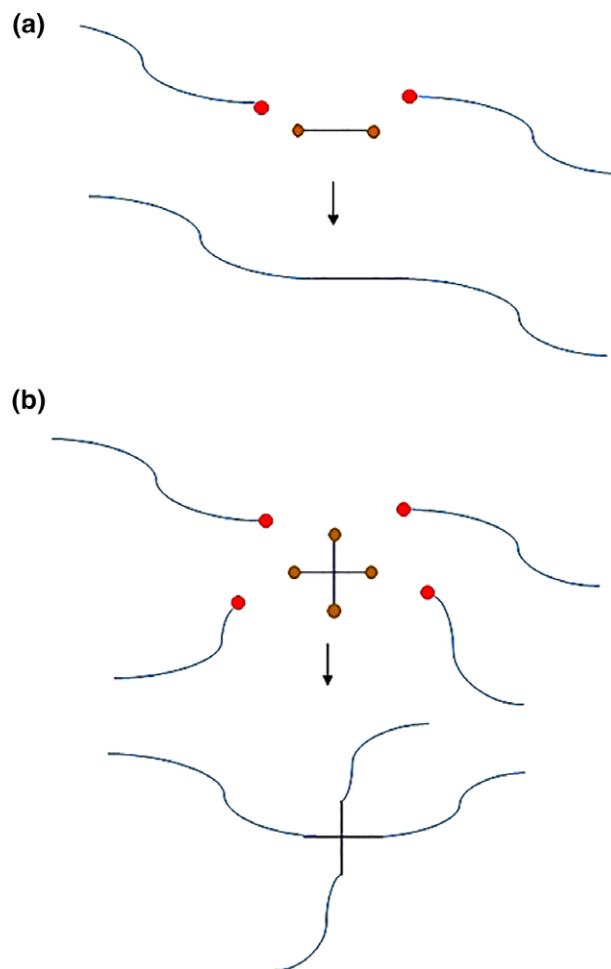


Figure 3. (a) Schematic of linear extension of polymer chains using a bifunctional chain extender; (b) schematic of branching of polymer chains using a multifunctional chain extender. [Color figure can be viewed at wileyonlinelibrary.com]

empirical observations in blow-molding and foam-extrusion trials to estimate melt strength, and no measurements of this property have been reported.

The effect of functionalized polymers, especially maleic anhydride (MA)-grafted polyolefins (PO-g-MA), known as coupling agents and compatibilizers, on the viscosity increase and branching of PET has not been studied. The reaction between PET and PO-g-MA and the formation of PO-g-PET linkages have been proposed in some investigations for compatibilizing PET blends.^{29–32} Zhang *et al.*³³ blended recycled PET (RPET) with MA grafted linear low density polyethylene (LLDPE-g-MA) and found the formation of PET-co-LLDPE-g-MA. They also used MA grafted styrene-ethylene-butylene-styrene (SEBS-g-MA) as compatibilizer in RPET/LLDPE blends and indicated that the MA groups of SEBS-g-MA reacted with the hydroxyl end groups of PET to synthesize PET-co-SEBS-g-MA.³⁴ Raffa *et al.*²⁴ mentioned the possibility of branched structure formation resulting from the reaction between PE-g-MA and PET (Figure 4).

In this research, low concentrations of polypropylene (PP)-g-MA were melt mixed with PET, and the effect of its content and

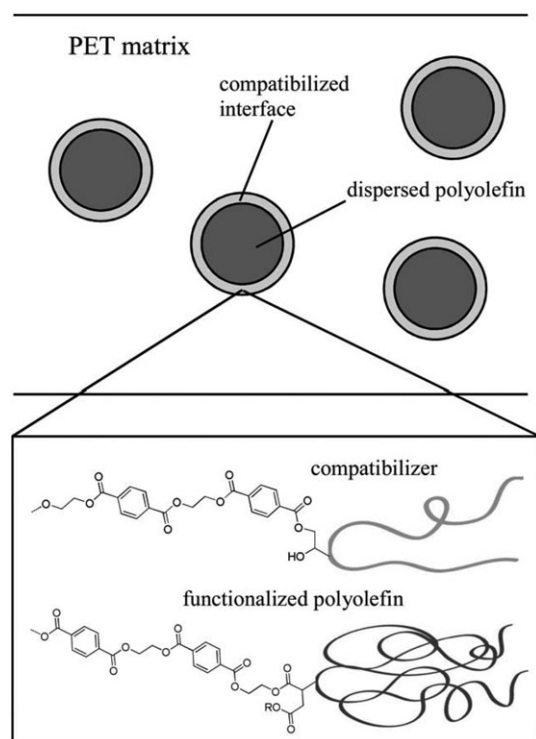


Figure 4. Representation of a PET/polyolefin compatibilized blend.¹⁹

grafting degree on the intrinsic viscosity and linear viscoelastic properties of PET were investigated.

EXPERIMENTAL

Experimental procedures were done in two parts. First, PP-g-MA was synthesized by the reactive melt-processing method, and then it was added to a PET melt with the aim of enhancing its rheological properties.

Materials

To synthesize the PP-g-MA, PP ZH525J with a melt flow index (MFI) of 3.1 g/10 min was supplied by Navid Zar Chimi Company (Tehran, Iran). Maleic anhydride was supplied by Daejung (Korea) as grafting monomer. Styrene (St) was supplied by Merck (Darmstadt, Germany) as a comonomer to enhance the grafting efficiency. Dicumyl peroxide (DCP) was supplied by Sigma-Aldrich (St. Louis, MO) as initiator. Irganox B225 was supplied by Ciba (Basel, Switzerland) as the antioxidant. PET TG641 with an intrinsic viscosity of 64 dL/g was supplied by Tondgooyan Petrochemical Company (Mahshahr, Iran).

Preparation of PP-g-MA

A reactive melt-processing method was used to functionalize PP by MA. First the internal mixer was heated to 190 °C, and then 35 g of PP was added to the chamber. After 2 min of mixing at a rate of 60 rpm, styrene in 1:1 molar ratio to MA was added to the compound by a syringe. One minute later, MA, and 30 s after that, dicumyl peroxide were added to the system. After approximately 8 min of total mixing time (more than five half-time of DCP at 190 °C), mixing was stopped. In order to prevent thermal oxidation, 0.25 wt % of Irganox B225 was used. Table I

Table I. Synthesized PP-g-MA Samples with Different Grafting Degrees

Sample	MA (phr)	Grafting degree
PP-g-MA1	5	1.67
PP-g-MA2	7	1.42
PP-g-MA3	9	1.18

represents the synthesized PP-g-MA samples, sorted by grafting degree.

Purification of PP-g-MA

First, 4 mg of the PP-g-MA sample was dissolved in refluxing xylene for 1 h. Then 250 mL of acetone was added to separate all ungrafted substitutes from PP-g-MA. Acetone can precipitate grafted and ungrafted PP [PP, PP-g-MA, PP-g-St, and PP-g-(SMA)] in the form of a white sediment. After filtering of the white sediment and washing with acetone several times, the sample was dried in a vacuum oven at 80 °C for 24 h.

Fourier Transform Infrared Spectroscopy

The anhydride content of the purified PP-g-MA samples was measured by Fourier transform infrared (FTIR) spectroscopy, recorded on a Vortex FTIR (Billerica, Massachusetts, US).

Melt Mixing of PET with PP-g-MA

In order to minimize the water content, all materials were dried in a vacuum oven for 24 h, at 120 °C for PET and 70 °C for PP-g-MA. The reactive melt-processing method was used for mixing PET with PP-g-MA. First, 40 g of PET was added to the internal mixer chamber, and after 4 min of mixing at 270 °C at 60 rpm, PP-g-MA and antioxidant were added. After 14 min of total mixing time, processing was stopped. The neat PET sample was also processed in the same conditions without adding any other component. Table II represents the neat (sample 1) and modified PET samples.

Intrinsic Viscosity Measurement

First, 25 mg of a sample was dissolved in 125 mL 50/50 wt % solution of phenol/tetrachloroethane at 120 °C for 2 h.⁸ The intrinsic viscosity of the solution was measured at 25 °C by a Ubbelohde viscometer and using the Solomon-Ciuta equation³⁵ where η_{sp} and η_{rel} are respectively specific viscosity and relative viscosity and c is the solution concentration:

$$[\eta] = [2(\eta_{sp} - \ln \eta_{rel})]^{0.5} / c \quad (1)$$

Table II. PET/MA Samples with Different PP-g-MA Contents

Sample	PP-g-MA type	PP-g-MA content (wt %)	Sample code
1	—	0	PET
2	PP-g-MA1	0.25	PET/MA1,25
3	PP-g-MA1	0.5	PET/MA1,5
4	PP-g-MA1	0.75	PET/MA1,75
5	PP-g-MA2	0.75	PET/MA2,75
6	PP-g-MA3	0.75	PET/MA3,75

Rheological Measurement

The rheological properties were investigated by small-amplitude oscillatory shear tests in the linear viscoelastic region (amplitude = 0.5%) and in frequency sweep mode using an MCR300 Anton Paar rheometer (Graz, Austria). The measurements were done under nitrogen atmosphere.

RESULTS AND DISCUSSION

PP-g-MA Characterization

During the functionalization of PP by MA using the reactive melt-processing method and in the presence of styrene as the comonomer, three main possibilities can be considered: the formation of grafted or ungrafted SMA,³⁶ the formation of grafted or ungrafted poly(maleic anhydride) (PMA),³⁷ or MA in grafted or ungrafted form.³⁷

Acetone can dissolve all ungrafted components, so after washing the samples with acetone and drying them in the vacuum oven, all ungrafted components would be eliminated and the grafted PP would remain. Figure 5 shows the FTIR spectra of the PP-g-MA samples, and Table III presents some common anhydride-containing groups and their corresponding FTIR peak positions.

There are no peaks at 1780 cm⁻¹ and 1792 cm⁻¹ in the FTIR spectra of the PP-g-MA samples. Based on Table III, these peaks are assigned to maleic anhydride and succinic anhydride, respectively. Therefore, one can assume there is a lack of single MA groups grafted onto PP chains. Meanwhile, a peak appearing around 1782–1784 cm⁻¹ is assigned to PMA or SMA.^{36,37}

The grafting degree of PP-g-MA samples was measured by FTIR using eqs. (2) and (3).³⁷

For *n*-octadecylsuccinic anhydride, we use

$$[\text{Anhydride}] = 51.3 \left(\frac{\text{Abs}1792\text{cm}^{-1}}{\text{Abs}1100\text{cm}^{-1}} \right) + 52.5 \left(\frac{\text{Abs}1715\text{cm}^{-1}}{\text{Abs}1100\text{cm}^{-1}} \right) \quad (2)$$

For PMA (or SMA), we use

$$[\text{Anhydride}] = 51.3 \left(\frac{\text{Abs}1784\text{cm}^{-1}}{\text{Abs}1100\text{cm}^{-1}} \right) + 52.5 \left(\frac{\text{Abs}1715\text{cm}^{-1}}{\text{Abs}1100\text{cm}^{-1}} \right) \quad (3)$$

Here, Abs 1100 cm⁻¹ is the polypropylene reference absorbance, Abs 1784 cm⁻¹ is the absorbance of PMA or SMA, Abs 1792 cm⁻¹ is the absorbance of the succinic anhydride symmetric C=O stretch, and Abs 1715 cm⁻¹ is the absorbance of the carboxylic acid C=O stretch. Only the first term of eq. (3) has a nonzero value, which represents the amount of MA grafted onto PP in the presence of styrene.

In some studies, the possibility of PMA formation has been ignored,³⁸ while in some others the polymerization of MA groups and grafting of PMA onto PP chains was reported.³⁷ However, in the presence of styrene, the most probable reaction is the formation of SMA.³⁶ So as a logical assumption, the PMA formation possibility can be ignored. Based on the above, the functionalized samples named as PP-g-MA could be considered as polypropylene chains grafted by SMA with possibly negligible amounts of MA.

MA-containing groups could be grafted along the chain or at the chain ends of polypropylene, considering that the chain end grafting is more probable.³⁷ Figure 6 is a schematic of grafting of MA-containing groups onto PP chains.

PET/MA Characterization

Intrinsic Viscosity. Intrinsic viscosity (IV) is a function of molecular weight (MW). Molecular weight enhancement with no introduced branches leads to an increase in intrinsic viscosity. The linear polymer regularly has a higher intrinsic viscosity than the corresponding branched one with the same molecular weight. This happens because the linear macromolecule has larger hydrodynamic dimensions in solution compared to the corresponding branched one with the same molecular weight.³⁵ In contrast, many studies reported an IV increase during the branching of the polymer.^{12,18,35} Table IV represents the IV of neat and modified PET samples and demonstrates the increase in modified ones. In samples 2, 3, and 4, which were all modified by PP-g-MA1, the IV increased with PP-g-MA concentration. In samples 4, 5, and 6, which were modified by the same concentration of PP-g-MA1, PP-g-MA2, and PP-g-MA3, respectively, the IV increased with the grafting degree of PP-g-MA. The highest IV value belongs to sample 4, due to modification by the highest content of PP-g-MA with the highest grafting degree. The IV of unprocessed neat PET (not presented in Table IV) was also measured (0.65 dL/g). This value decreased to 0.54 dL/g after processing (sample 1), due to MW reduction caused by degradation reactions. The IVs of modified PET samples are between these two numbers, showing that the addition of PP-g-MA to PET suppresses the effects of degradation. The IV increase of the PET/MA samples and its dependence on the amount of MA groups may be due to the reactions between PP-g-MA and PET.

Linear Viscoelastic Properties. Storage and loss moduli.

Rheological properties are very sensitive to molecular structure. In this study, the viscoelastic properties of samples were studied by shear oscillatory measurements in the linear viscoelastic region. Figure 7(a,b) shows the storage (*G'*) and loss (*G''*) moduli versus frequency (*ω*) at 265 °C for samples 1 and 4, respectively. Measurements at frequencies lower than 0.1 rad/s were not possible due to the degradation of PET after a long time. In Figure 7(a) there is no crossover point in the measured frequency range. Actually, the crossover appears at frequencies higher than the frequency range used in this test; in Figure 7(b), the crossover appears at a certain frequency between 100 and 1000 rad/s. The shifting of the crossover point to lower frequencies for modified PET means higher elasticity, and it can be an indication of reactions occurring between PET and PP-g-MA. Similar diagrams were observed for samples 2 and 3, with a crossover point between 100 and 1000 rad/s.

The *G'*–*ω* diagrams of samples 1 to 4 are shown in Figure 8. At low frequencies, the PET/MA samples show a bit higher value of *G'* than the neat PET, and the difference is more noticeable with increasing PP-g-MA content. At low frequencies, molecules with higher molecular weight determine the rheological behavior of the polymer. In other words, the MW of polymer plays the major role in the terminal zone.³⁹ Thus the higher *G'* values in the PET/MA samples indicate the formation of molecular structures with higher MW. This demonstrates the chain-extending effect of

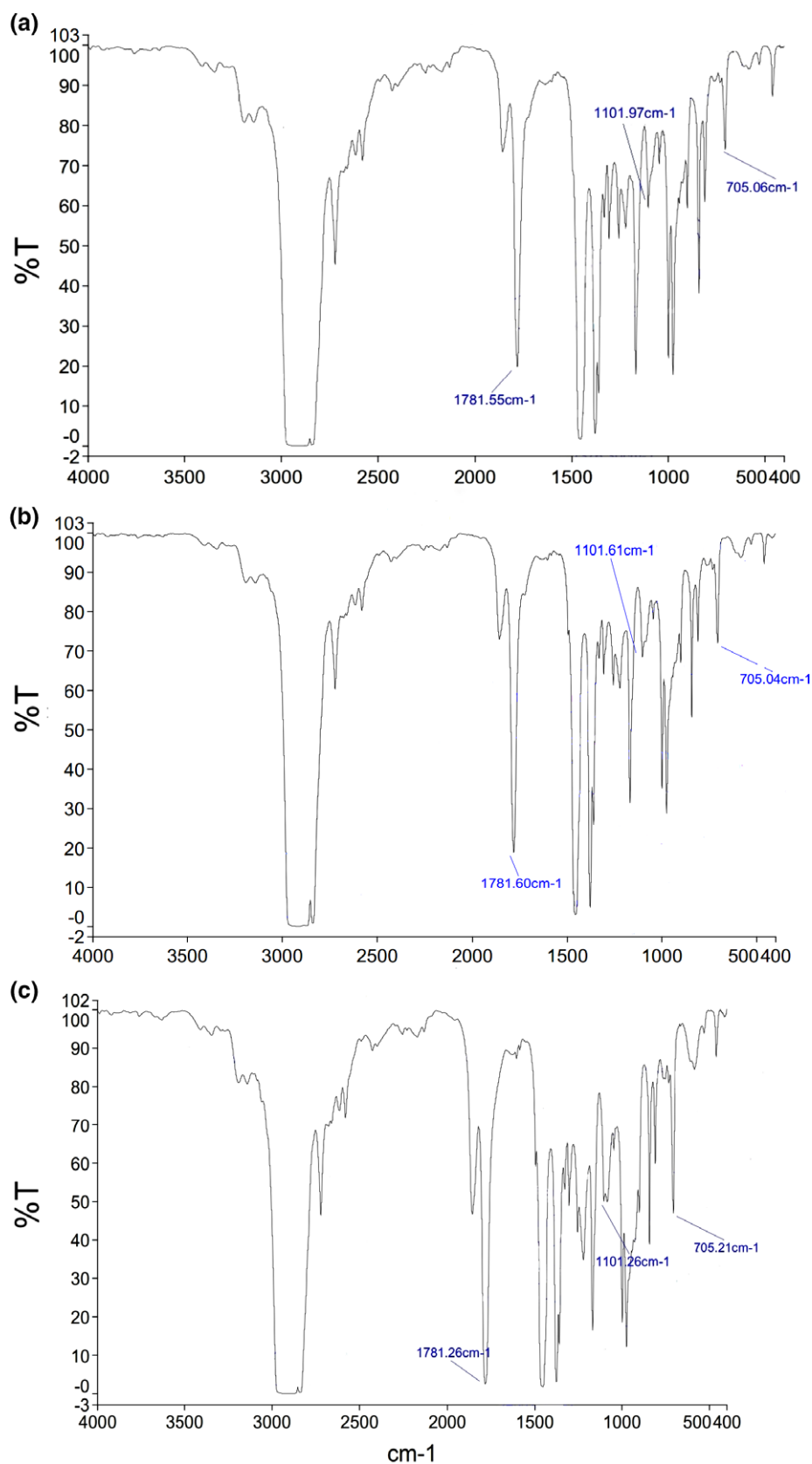
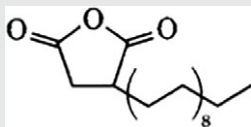
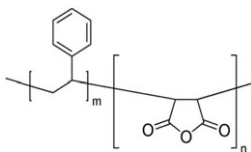
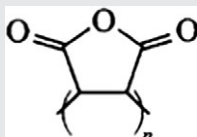
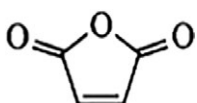


Figure 5. FTIR spectra of (a) PP-g-MA1, (b) PP-g-MA2, and (c) PP-g-MA3. [Color figure can be viewed at wileyonlinelibrary.com]

Table III. Molecular Structure and FTIR Peak Position of Anhydride-Containing Chemicals

Component	Chemical structure	Peak position	Reference
<i>n</i> -Octadecylsuccinic anhydride		1792	37
Poly(styrene-maleic anhydride)		1782	36
Poly(maleic anhydride)		1782-1784	36, 37
Maleic anhydride		1780	37

PP-g-MA on PET, although the difference is not significant. At high frequencies, the difference between the diagrams for neat PET and the PET/MA samples are more noticeable. The reason will be discussed in the next sections.

Figure 9 represents the G' - ω diagram of samples 4 to 6 (PET/MA samples with equal amounts of PP-g-MA1, PP-g-MA2, and PP-g-MA3, respectively). The plots for samples 5 and 6 are matched, while sample 4 shows higher G' values. Higher amounts of anhydride groups in PP-g-MA lead to more reactions with hydroxyl end groups on PET and consequently higher elastic behavior.

Complex viscosity. As shown in Figure 10, at 0.1–100 rad/s, all samples approximately show a Newtonian behavior, and the complex viscosity is almost constant regardless of frequency. In this region, the complex viscosity of the modified samples is higher than in neat PET, and the difference will be more significant with increasing PP-g-MA content. The higher values of complex viscosity of the modified samples can be due to the formation of molecules with higher MW due to the reaction between PET and PP-g-MA, as mentioned before. Higher PP-g-MA content leads to more reactions and consequently higher values of complex viscosity.

**Figure 6.** Schematic of grafting of MA-containing groups onto PP chains. [Color figure can be viewed at wileyonlinelibrary.com]

Beyond a specific frequency around 100 rad/s, the complex viscosity of neat PET decreases slightly, which could be due to the degradation of PET chains (Figures 1 and 2). In contrast, the diagrams of PET/MA samples show an unusual increase in this region. One can conclude that the MA groups are responsible for this phenomenon. Besides, the onset of viscosity increase in the PET/MA samples matches the onset of the decrease in viscosity of neat PET. Based on the above, it may be concluded that the possible reaction between MA groups and the functional groups resulting from PET degradation (in N_2 atmosphere) causes the unusual viscosity upturn in the PET/MA samples.

In the absence of oxygen, nonradical thermal degradation²⁷ and hydrolysis⁴⁰ are the only possible degradation reactions of PET. Regarding hydrolysis, it is noteworthy that even in well-dried samples, hydrolysis may happen.⁴⁰ As represented in Figure 2, hydrolysis of PET forms the shorter chains containing the hydroxyl groups. Colin and Verdu²⁷ proposed that the anhydrides are highly sensitive to hydrolysis and can react in transesterification processes; in certain conditions, they can even form branches. Therefore, we propose that the transesterification reactions between the anhydride groups of PP-g-MA and the hydroxyl groups formed during the hydrolysis of PET are responsible for the significant complex viscosity upturn that appears in the η^* - ω diagrams of PET/MA samples.

Table IV. Intrinsic Viscosity of Neat PET and PET/MA Samples

Sample	Sample code	Intrinsic viscosity
1	PET	0.543
2	PET/MA1,25	0.572
3	PET/MA1,5	0.608
4	PET/MA1,75	0.615
5	PET/MA2,75	0.579
6	PET/MA3,75	0.566

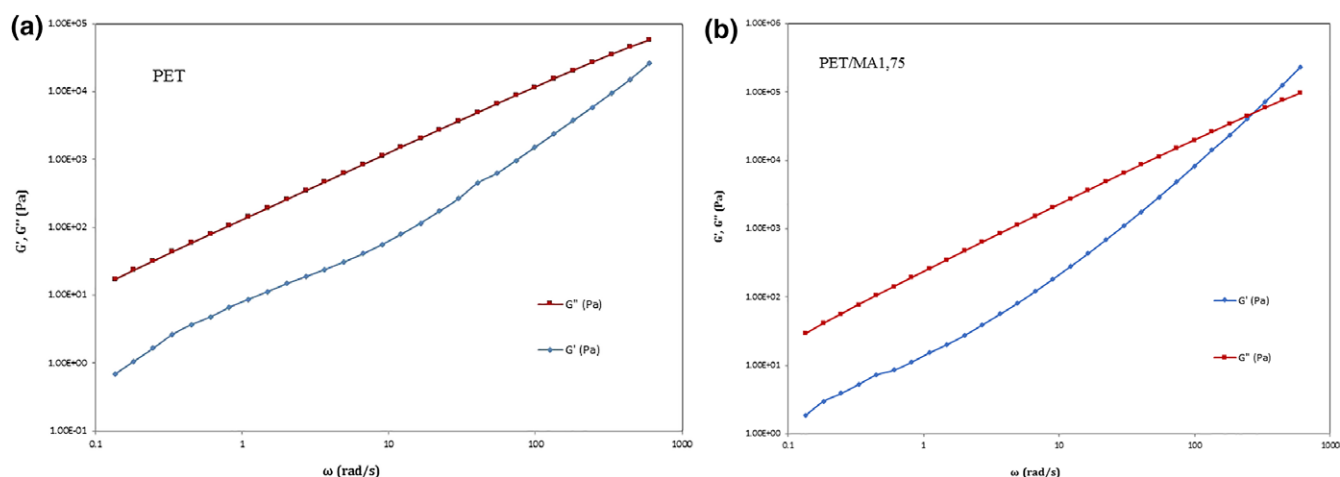


Figure 7. G' and G'' versus frequency (ω) at 265 °C for (a) neat PET and (b) PET/MA1,75. [Color figure can be viewed at wileyonlinelibrary.com]

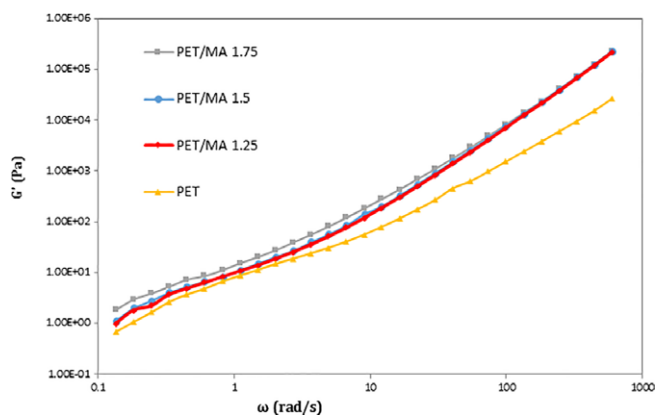


Figure 8. G' - ω diagram of samples 1 to 4 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

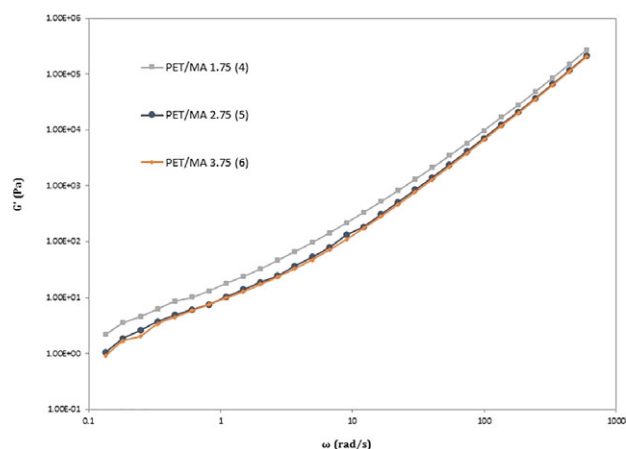


Figure 9. G' - ω diagram of samples 4 to 6 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

Given that no shear thinning behavior is observed in any sample diagram, it may be concluded that there are no branched structures in the PET/MA samples. Actually, as mentioned before, during MA grafting of PP, chain end grafting is much more probable than grafting along the chain. Furthermore, MA

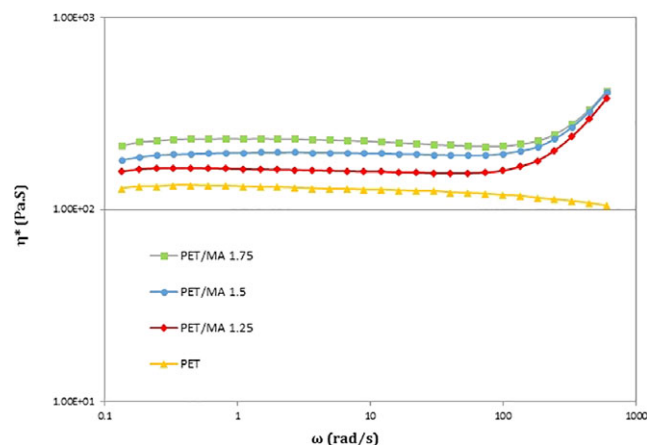


Figure 10. η^* - ω diagram of samples 1 to 4 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

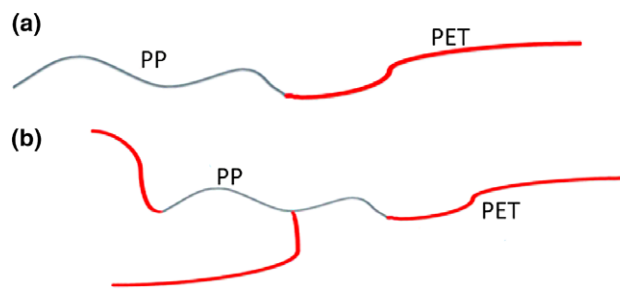


Figure 11. Schematics of possible structures during melt mixing of PET with PP-g-MA: (a) copolymer with linear structure, (b) copolymer with branched structure. [Color figure can be viewed at wileyonlinelibrary.com]

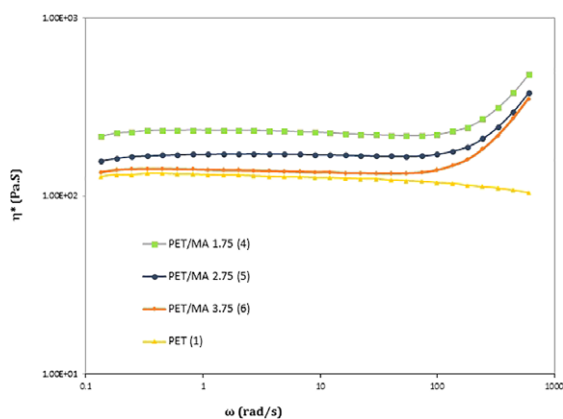


Figure 12. η^* – ω diagram of neat PET and samples 4 to 6 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

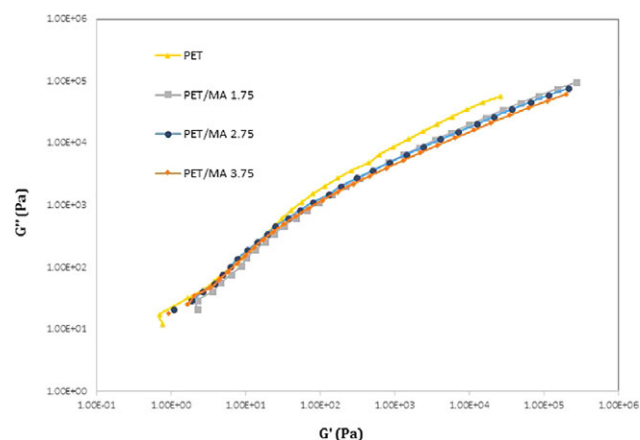


Figure 15. Cole–Cole diagram of neat PET and samples 4 to 6 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

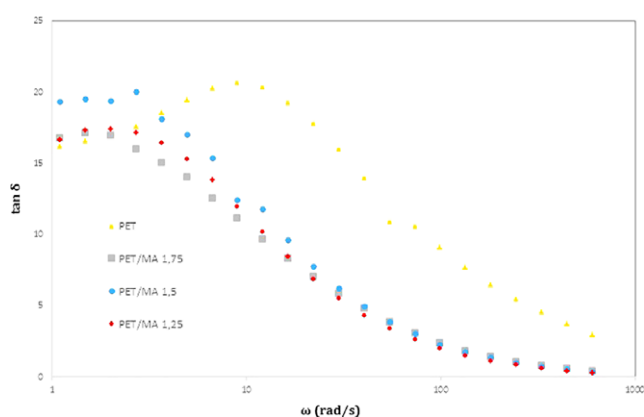


Figure 13. $\tan \delta$ – ω diagram of samples 1 to 4 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

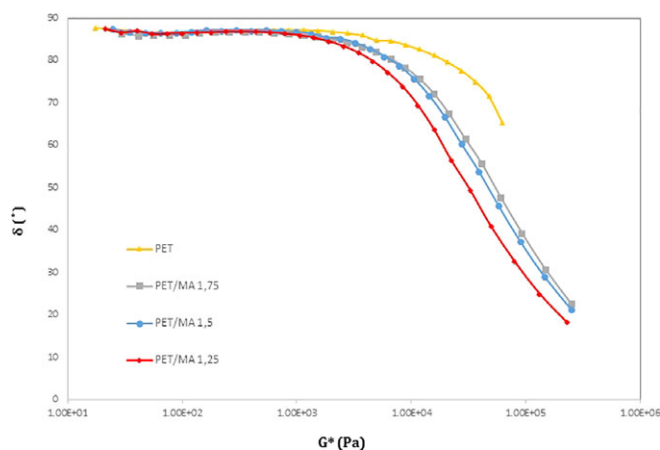


Figure 16. Van Gurp–Palmen diagram of samples 1 to 4 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

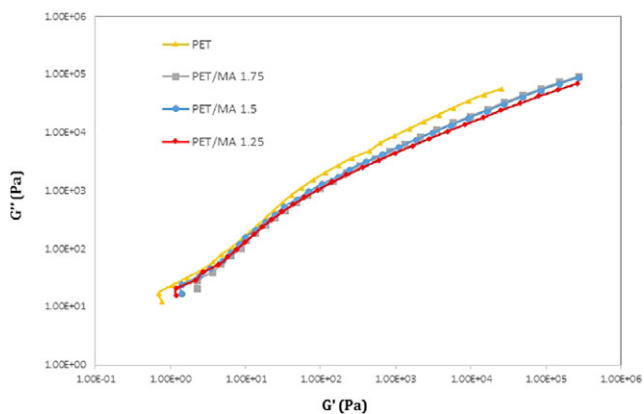


Figure 14. Cole–Cole diagram of samples 1 to 4 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

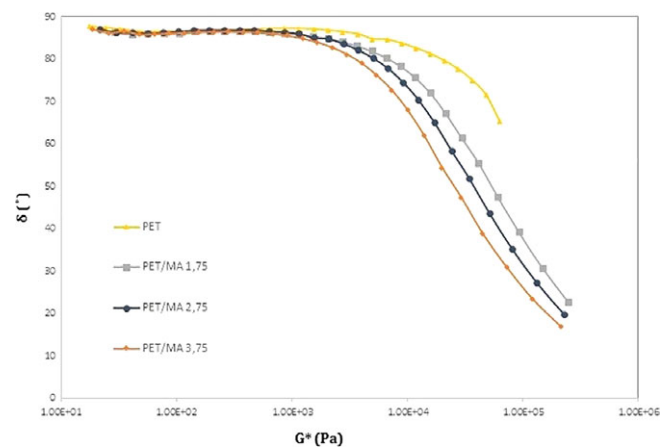


Figure 17. Van Gurp–Palmen diagram of neat PET and samples 4 to 6 at 265 °C. [Color figure can be viewed at wileyonlinelibrary.com]

groups grafted along the chain may even be encapsulated by PP chains, which will reduce the chance of contact between the MA groups of PP-g-MA and the hydroxyl end groups of PET.⁷ Thus, during PET/PP-g-MA reaction and the formation

of PET-co-PP-g-MA, the possibility of formation of a linear structure [Figure 11(a)] is higher than that of a branched one [Figure 11(b)]. This correlates with the results obtained from the intrinsic viscosity measurements.

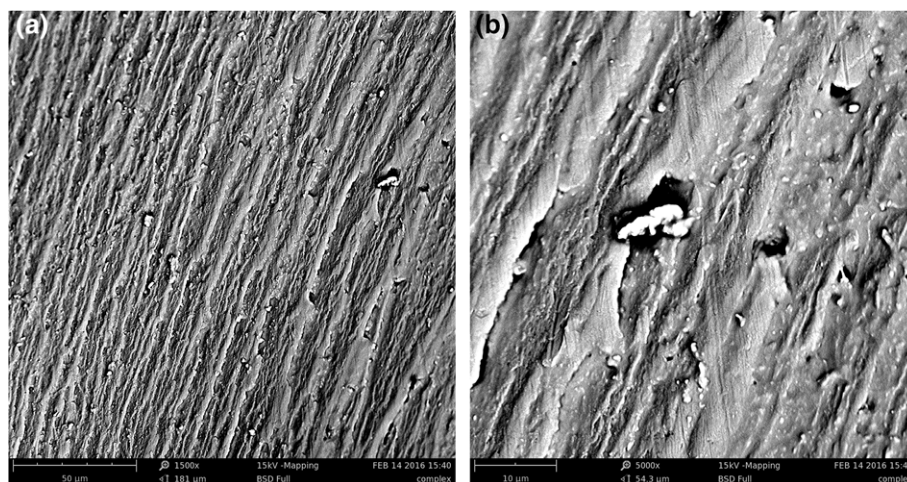


Figure 18. SEM micrographs of PET/MA1,75 before rheometry: (a) 1500 $\times$, (b) 5000 $\times$.

Figure 12 shows the variations of complex viscosity versus frequency for neat PET and samples 4 to 6. Complex viscosity increases with the increase in grafting degree of PP-g-MA. The effect of grafting degree on complex viscosity is similar to the effect of PP-g-MA content. The viscosity upturn can also be observed in these samples.

Damping factor. Damping factor ($\tan \delta$), representing the ratio of viscous to elastic responses of a polymer melt, is one of the parameters that can be used to describe the polymer rheological behavior and consequently to characterize the molecular structure. Figure 13 shows the dependence of $\tan \delta$ on frequency for samples 1 to 4. In a broad range of middle frequencies, the $\tan \delta$ values of PET/MA samples are lower than that of neat PET, and at high frequencies their dependence on frequency is lower. The former is due to the formation of linear PET-co-PP-g-MA, and the latter could be a result of the formation of complex linear/branched structures due to the transesterification reactions (Figure 11).

There is no significant difference between the modified samples, but at some frequencies, sample 4 shows the highest values, due to its highest PP-g-MA content with highest grafting degree. The diagrams of samples 5 and 6 (not presented here) are just like that of samples 2 and 3, and the grafting degree of PP-g-MA has an effect similar to PP-g-MA content.

Cole-Cole diagram. Figure 14 shows the modified Cole-Cole diagrams of samples 1 to 4. At low values of G' , there is no difference between diagrams, while at high values, where the elastic portion plays the principal role in polymer behavior, the slopes of the modified-PET diagrams differ from the neat PET. Deviation from linear PET indicates the formation of new molecular structures during the transesterification reactions in modified samples (Figure 11).

Figure 15 shows the modified Cole-Cole diagrams of neat PET and samples 4 to 6. Similar to Figure 14, the slopes of the plots for modified samples differ from the neat PET. In both figures, there is no significant difference between modified samples.

Van Gorp–Palmen plot. A Van Gorp–Palmen plot is independent of MW, while it is sensitive to the long chain branches and molecular weight distribution. The closer the diagram is to $\delta = 90^\circ$ indicates the linearity of the polymer. Figure 16 and 17 show the Van Gorp–Palmen plots of samples. At high values of G^* , modified

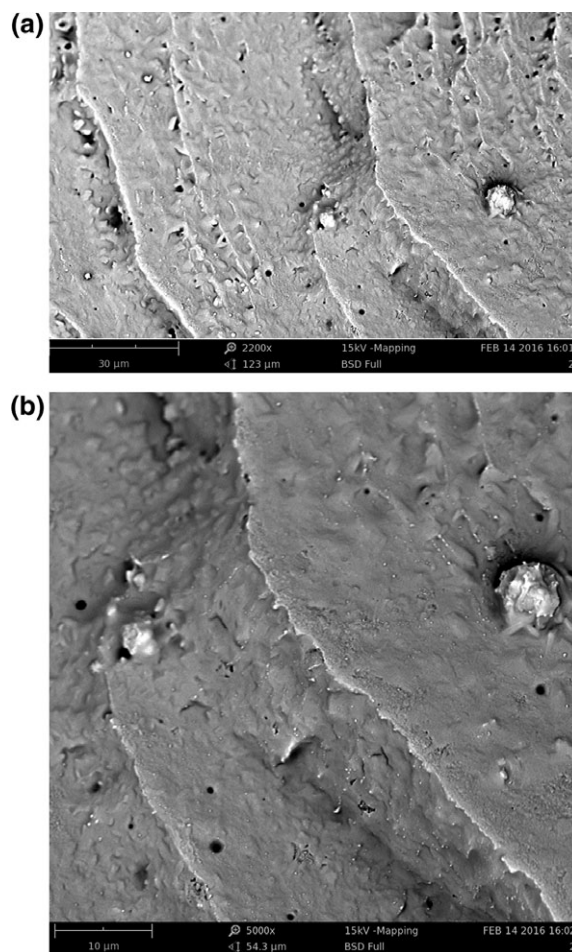


Figure 19. SEM micrographs of PET/MA1,75 after rheometry: (a) 2500 $\times$, (b) 5000 $\times$.

Table V. Thermal Properties of PET and PET/MA1,75 Samples

Sample	Sample code	T_g (°C)	T_c (°C)	T_m (°C)	ΔH_c (J/kg)	ΔH_m (J/kg)	Degree of crystallization
1	PET	83.9	193.8	256.6	40.5	38	31.4
4	PET/MA1,25	81.2	203.8	254.8	41.2	47.1	29.4

samples show a significant deviation from the neat PET. Deviation from the linear polymer demonstrates a broadening of the molecular weight distribution or branching of the polymer during transesterification reactions (Figure 11). In both figures, the differences between the modified samples are not significant.

Scanning Electron Microscopy

Figures 18 and 19 show scanning electron microscopy (SEM) micrographs of PET/MA1,75 before and after rheometry tests, respectively. PP-g-MA particles are well dispersed in the PET matrix, and as seen, their adhesion to PET is enhanced after rheometry. Actually, in Figure 19 the morphology of the sample is more homogenous compared to Figure 18, indicating the occurrence of a transesterification reaction between PET and PP-g-MA.

Differential Scanning Calorimetry

Thermal properties like melting point (T_m), glass-transition temperature (T_g), and crystallization temperature (T_c) may be significantly influenced by chain extension reactions. Table V shows the differential scanning calorimetry (DSC) test results for the PET and PET/MA1,75 samples. The T_g of the modified sample showed a decrease of 2.7 °C, which may be due to the increased free volume resulting from mobility of polypropylene segments. The higher values obtained for heat and enthalpy of crystallization demonstrate that in the modified sample, the rearrangement of polymer chains needs more energy than the unmodified one, which probably is due to the disordered molecular structure resulting from chain extension reactions. The decrease in crystallization degree from 31.4 to 29.4 also confirms the former explanation. The lower value of melting point in the modified sample indicates smaller crystals and more defects in lamellae.

CONCLUSIONS

Reactive melt processing of PET with PP-g-MA influences the intrinsic viscosity and rheological properties of PET. The addition of PP-g-MA leads to an increase in intrinsic viscosity, which confirms the formation of new molecular structures due to the reaction between PP-g-MA and PET. Rheological measurements show the shifting of the crossover point to lower frequencies for PET/MA samples demonstrating higher elasticity. The increase in PP-g-MA content or grafting degree leads to increasing intrinsic and complex viscosities and enhances the elastic behavior. Damping factor, Cole–Cole, and Van Gurp–Palmen plots indicate the formation of PET-co-PP-g-MA with two proposed linear and branched structures. Given that in the PP-g-MA structure, the probability of the presence of MA groups at the chain ends is higher than for their presence along the chain, and given that no shear thinning behavior was observed for any sample, it may be concluded that branching is not widespread. An unusual complex viscosity upturn was observed at high frequencies in $\eta^*-\omega$ diagrams for all PET/MA samples. We propose that the

transesterification reactions between the anhydride groups of PP-g-MA and the hydroxyl groups formed during hydrolysis of PET are the reason for this phenomenon.

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